

Extended weak order

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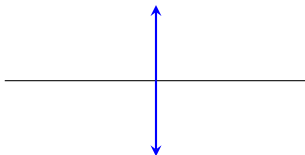
July 6, 2026

Root systems and Coxeter groups

Root systems

Let V be a real vector space with symmetric bilinear form $(-, -)$.
If $\alpha \in V$ has $(\alpha, \alpha) > 0$, then the **reflection over α** is

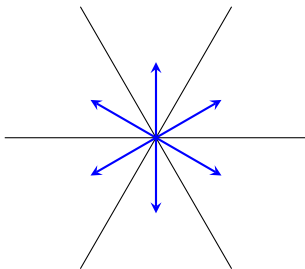
$$t_\alpha : \mathbf{v} \mapsto \mathbf{v} - 2 \frac{(\alpha, \mathbf{v})}{(\alpha, \alpha)} \alpha$$



Root systems

A **finite root system** is a finite collection of vectors (called **roots**) so that

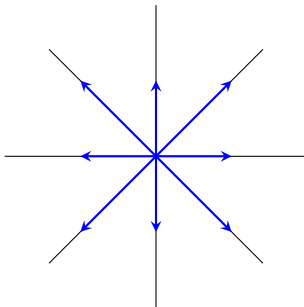
- ▶ Reflecting any root over any other root gives another root
- ▶ Any line contains at most two roots



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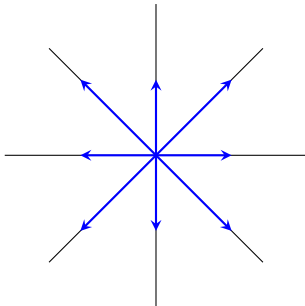
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The collection of hyperplanes dual to the roots is called the **Coxeter arrangement**

General root systems

Definition

A (real) **root system** is a set $\Phi \subseteq V$ of **roots** so that:

- ▶ Each $\alpha \in \Phi$ has $(\alpha, \alpha) > 0$ and $t_\alpha \Phi = \Phi$, and

General root systems

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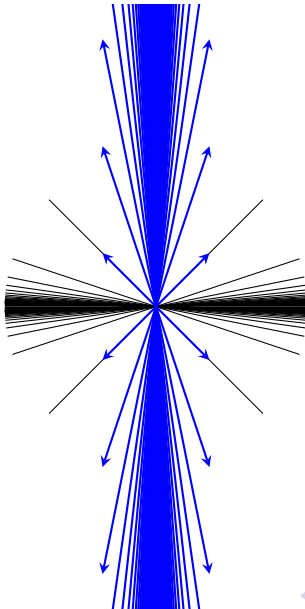
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Let W be the **Weyl group** of Φ , which is the subgroup of $GL(V)$ generated by $\{t_\alpha \mid \alpha \in V\}$. Then we further require:

- ▶ There exists a linearly independent set $\Pi \subseteq \Phi$ so that:
 - ▶ $\Phi = \bigcup_{w \in W} w\Pi$
 - ▶ $\Phi \subseteq \text{cone}(\Pi) \cup \text{cone}(-\Pi)$
- ▶ For any $\alpha \in \Phi$, $\mathbb{R}\alpha \cap \Phi = \{\pm\alpha\}$

An infinite root system: $\tilde{A}_1$



Root systems and Coxeter groups

The Weyl group of Φ is the group W generated by reflections over roots

Theorem

The Weyl groups of root systems are exactly the **Coxeter groups**: groups with a set of generators $S = \{s_i\}_{i \in I}$ so that

$$W \cong \langle S \mid (s_i s_j)^{m_{ij}} = 1, \ i, j \in I \rangle.$$

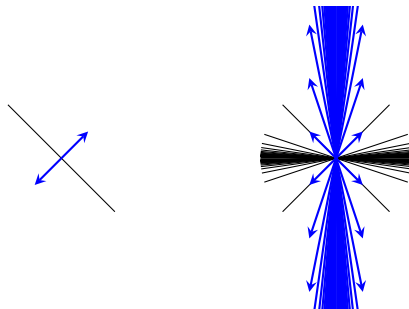
Here $m_{ij} \in \mathbb{Z}_{\geq 2} \cup \{\infty\}$, and we require $m_{ij} = m_{ji}$ and $m_{ii} = 1$.

Finite Coxeter groups are classified by families $A_n, B_n = C_n, D_n, I_2(m)$, and exceptional groups $E_6, E_7, E_8, F_4, H_3, H_4$.

Affine root systems

Let Φ be a **crystallographic** finite root system: meaning there exists a basis of roots $\Pi = \{\alpha_1, \dots, \alpha_n\}$ so that $\Phi \subseteq \mathbb{Z}\Pi$

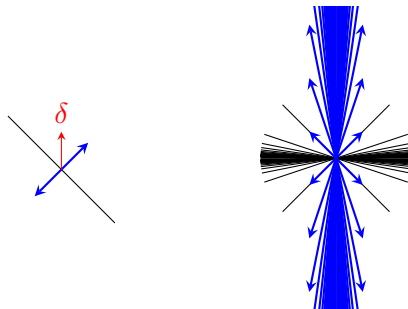
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Affine root systems

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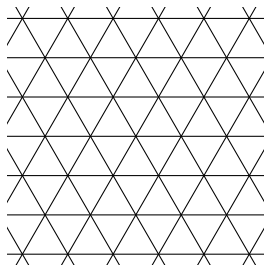
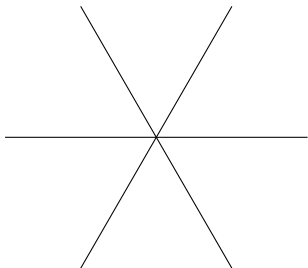


Example

The symmetric group S_n is a Coxeter group. Its root system is called the **type A_{n-1} root system**:

$$\Phi = \{\mathbf{e}_i - \mathbf{e}_j \mid i \neq j, i, j \in [n]\}$$

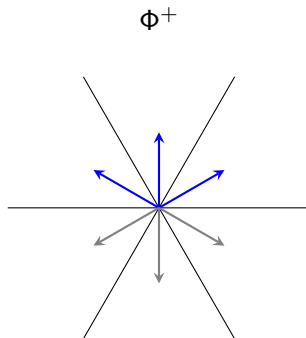
The corresponding affine root system is called the **type $\tilde{A}_{n-1}$ root system**. Its Weyl group is the **affine symmetric group $\tilde{S}_n$** .



Weak order

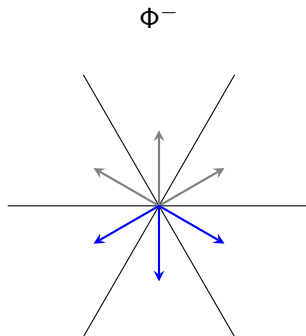
Positive roots

Root systems can be split into **positive roots** and **negative roots**



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Separable sets of roots

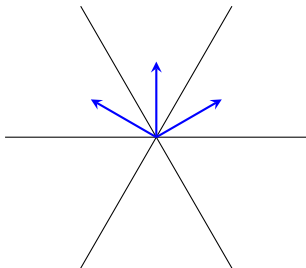
A **separable set** of positive roots is $B \subseteq \Phi^+$ so there is a hyperplane separating B from $\Phi^+ \setminus B$.

Equivalently, there is a linear map $f : V \rightarrow \mathbb{R}$ so that $f(B) < 0$ and $f(\Phi^+ \setminus B) > 0$.

Separable sets correspond to regions of the Coxeter arrangement.

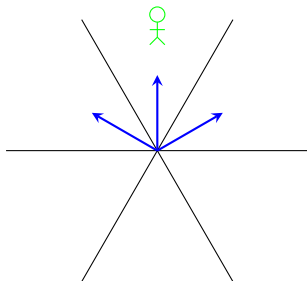
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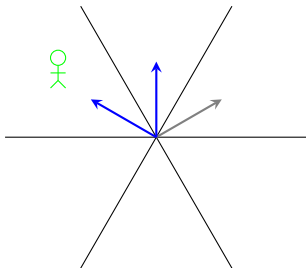
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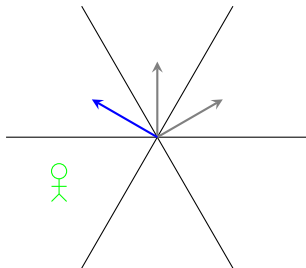
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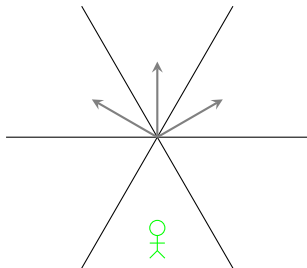
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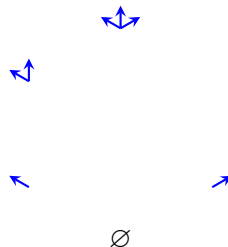
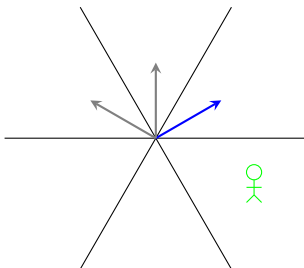
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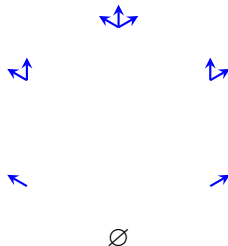
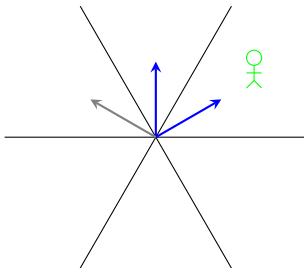
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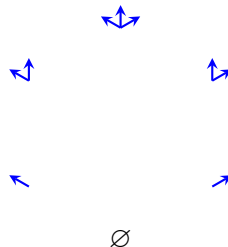
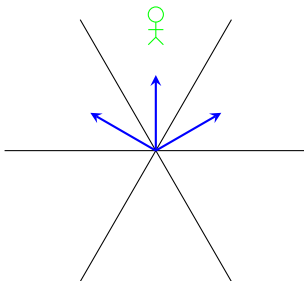
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Weak order

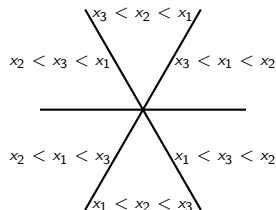
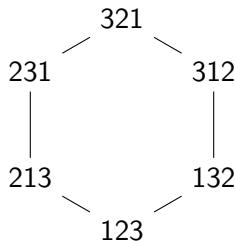
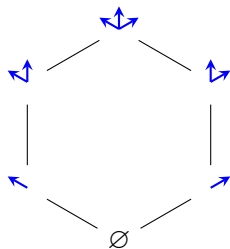
There is a separable set associated to each $w \in W$.

Definition

The **inversion set** of $w \in W$ is

$$\text{Inv}(w) := \{\alpha \in \Phi^+ \mid w^{-1}\alpha \in \Phi^-\}.$$

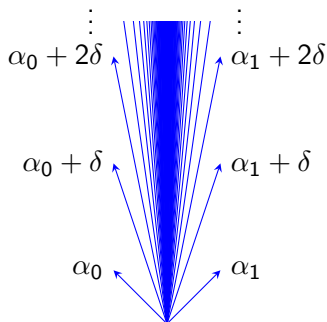
The **weak order** on W puts $w_1 \leq w_2$ if $\text{Inv}(w_1) \subseteq \text{Inv}(w_2)$.



Roots for $\tilde{A}_1$

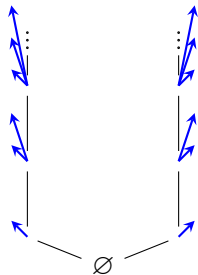
Let $\Phi = \{\pm\alpha_1\}$ be the A_1 root system. Set $\alpha_0 := \delta - \alpha_1$. The positive roots of the type $\tilde{A}_1$ affine root system are

$$\Phi^+ = \{\alpha_0 + n\delta, \alpha_1 + n\delta \mid n \in \mathbb{Z}_{\geq 0}\}.$$

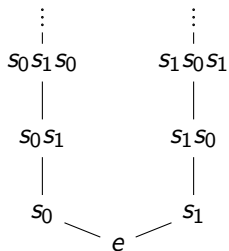


Weak order on $\tilde{S}_2$

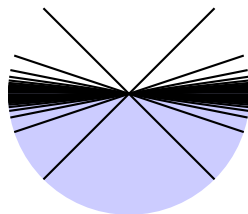
Inversion sets for $\tilde{S}_2$



Weak order on $\tilde{S}_2$



Coxeter arrangement

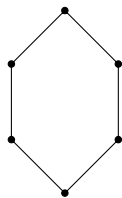


The lattice property

Definition

A partial order $\leq$ on $\mathcal{X}$ is a **complete lattice** if, for any collection $\{x_i\}_{i \in I} \subseteq \mathcal{X}$, the following two conditions hold:

1. There is a least upper bound $\bigvee_{i \in I} x_i$ of $\{x_i\}_{i \in I}$ in $\mathcal{X}$ (the **join** of $\{x_i\}_{i \in I}$)
2. There is a greatest lower bound $\bigwedge_{i \in I} x_i$ of $\{x_i\}_{i \in I}$ in $\mathcal{X}$ (the **meet** of $\{x_i\}_{i \in I}$)



The lattice property

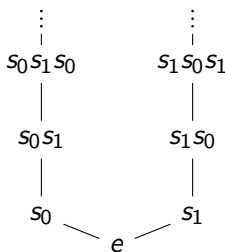
Theorem (Björner, Björner–Edelman–Ziegler, ...)

If W is a finite Coxeter group, then the weak order on W is a complete lattice.

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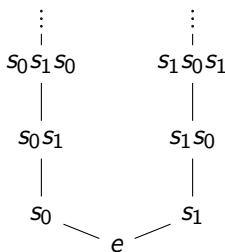


This fails when W is infinite.

The lattice property

Theorem (Björner, Björner–Edelman–Ziegler, ...)

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This fails when W is infinite.

Interpretation: weak order on infinite W has missing elements.

What are these missing elements?

Biclosed sets

Biclosed sets

A set $B \subseteq \Phi^+$ positive roots is **biclosed** if it satisfies the following two properties for all $\alpha, \beta, \gamma \in \Phi^+$ such that $\gamma = a\alpha + b\beta$ with $a, b \geq 0$:

(Closed) If $\alpha \in B$ and $\beta \in B$, then $\gamma \in B$.

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Example: Biclosed sets for S_3

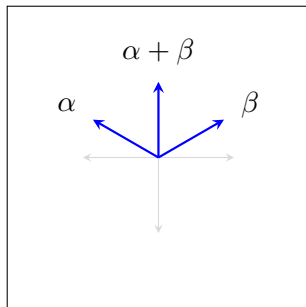
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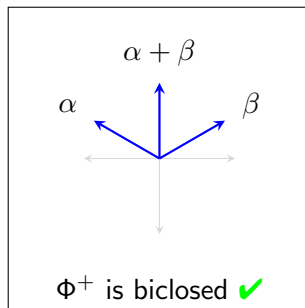
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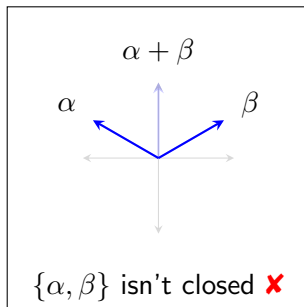
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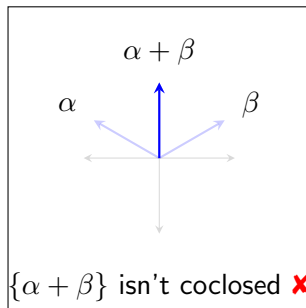
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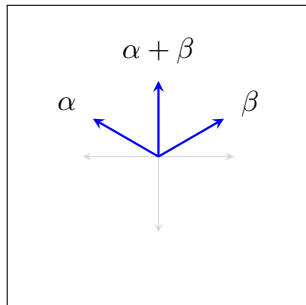
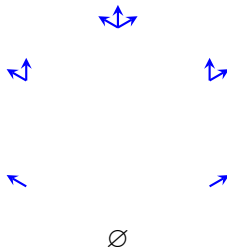
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Biclosed sets in
 Φ^+ :



Example: Biclosed sets for S_3

Definition

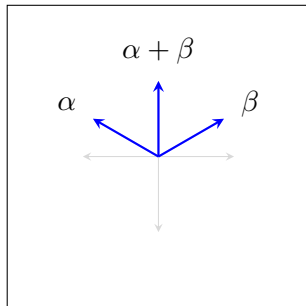
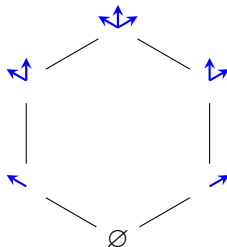
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Biclosed sets in Φ^+ :



Biclosed sets are locally separable

In dimension 2, the biclosed sets coincide with the separable sets.

Fact

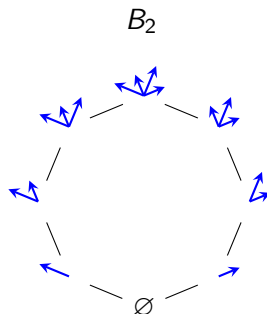
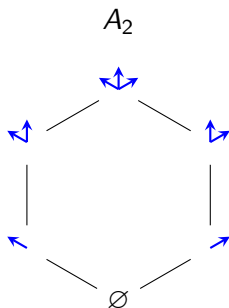
$B \subseteq \Phi^+$ is biclosed iff for any 2-dimensional space V' , the restriction $B \cap V'$ is a separable set in $\Phi^+ \cap V'$.

Extended weak order

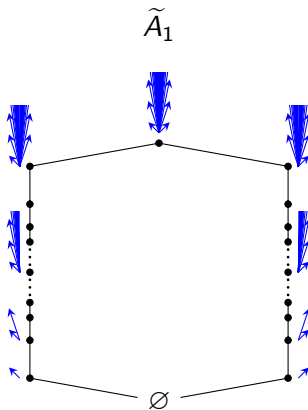
Definition

If Φ^+ is the set of positive roots for a Coxeter group W , then the poset $\text{Bic}(\Phi^+)$ consisting of biclosed sets, ordered by containment, is called the **extended weak order** of W .

Extended weak order



Extended weak order

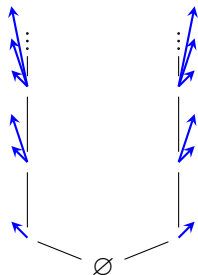


Why do biclosed sets extend weak order?

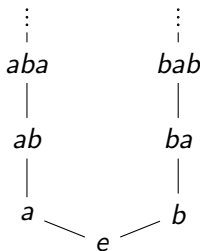
Theorem (Dyer, Cellini–Papi, Ito, ...)

If Φ is a root system, then the inversion sets $\text{Inv}(w)$ for $w \in W$ are exactly the *finite* biclosed sets in Φ^+ . These correspond to regions of the Coxeter arrangement in the **Tits cone**.

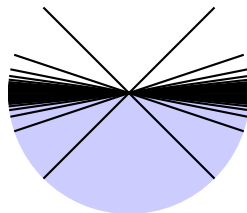
Finite biclosed sets in
 Φ^+



Weak order on $\tilde{S}_2$

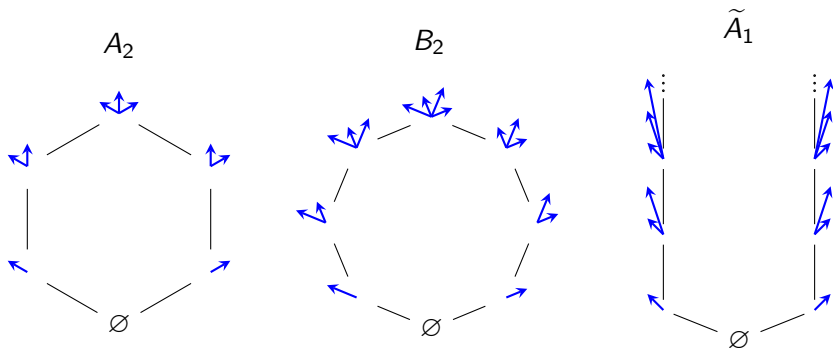


Coxeter arrangement
for Φ



Why do biclosed sets extend weak order?

By the theorem, the collection of finite biclosed sets (an order ideal in extended weak order) is isomorphic to weak order on W



History

- ▶ Finite separable sets well-studied (Coxeter complex)
- ▶ Other separable sets also studied:
 - ▶ Borel subalgebras of Kac–Moody algebras
 - ▶ stability conditions for quiver representations
- ▶ Biclosed sets introduced by Matthew Dyer and Tom Edgar:
 - ▶ reflection orders
 - ▶ twisted Bruhat orders
 - ▶ twisted Kazhdan–Lusztig polynomials

Dyer's main conjectures

Let W be any Coxeter group.

Dyer's Conjecture A

If $B_1 \triangleleft B_2$ is a cover relation in extended weak order, then

$$|B_2 \setminus B_1| = 1.$$

Dyer's Conjecture B

The extended weak order of W is a complete lattice.

Why not use separable sets?

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Fact

If $B_1 \triangleleft B_2$ is a cover relation in extended weak order, and both are separable sets, then $|B_2 \setminus B_1| = 1$.

Question: Why use biclosed sets instead of separable sets then?

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If $B_1 \triangleleft B_2$ is a cover relation in extended weak order, and both are separable sets, then $|B_2 \setminus B_1| = 1$.

Question: Why use biclosed sets instead of separable sets then?

One answer: In any affine Coxeter group of rank ≥ 4 , the separable sets fail Dyer's Conjecture B: they still have missing elements!

Dyer's Conjecture A

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Theorem (Dyer, B–Speyer 2023)

Dyer's Conjecture A is true for affine Coxeter groups.

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Theorem (Dyer, B–Speyer 2023)

Dyer's Conjecture A is true for affine Coxeter groups.

Theorem (B–Defant–Hersh–McCammond–McConville–Speyer 2025, 2026+)

Dyer's Conjecture A is true for rank 3 Coxeter groups.

Dyer's Conjecture B

For any root system Φ , extended weak order is a complete lattice.

Dyer's Conjecture B

For any root system Φ , extended weak order is a complete lattice.

Theorem (Björner 1984)

The finite biclosed sets (i.e. the weak order) for any Coxeter group form a meet-semilattice.

Dyer's Conjecture B

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Clean arrangements

Definition

Let $\mathcal{A}$ be a hyperplane arrangement, and let X be the set of normal vectors to the hyperplanes pointing towards some region. (For instance, $\mathcal{A}$ is a Coxeter arrangement and X is the positive roots.)

We say that $\mathcal{A}$ is a **clean arrangement** if every biclosed set in X is a separable set.

Theorem (B–Speyer 2023)

Every “ideal subarrangement” of a finite or rank 3 affine Coxeter arrangement is a clean arrangement.

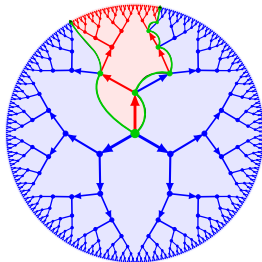
Theorem (Yoshinaga 2024)

Every finite $K(\pi, 1)$ arrangement is a clean arrangement.

Rank 3 cleanliness: snakes on a hyperbolic plane

Theorem (B–Defant–Hersh–McCammond–McConville–Speyer 2025, 2026+)

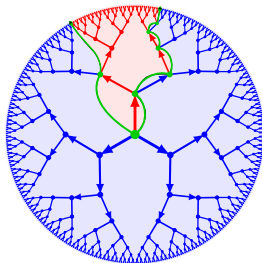
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Every rank 3 Coxeter arrangement is a clean arrangement.



Rank 4 affine Coxeter arrangements are not clean!

Biclosed sets for S_∞ and $\tilde{A}_{n-1}$

An example: S_∞

Let S_∞ be the group of permutations of $\mathbb{Z}$ fixing all but finitely many integers

S_∞ is a Coxeter group. What does extended weak order look like for it?

Observation

The weak order on permutations of $\mathbb{Z}$ does not form a complete lattice.

Example issue: Joining

$$\pi_0 = \dots, -3, -2, -1, \mathbf{0}, 1, 2, 3\dots$$

$$\pi_1 = \dots, -3, -2, \mathbf{0}, -1, 1, 2, 3\dots$$

$$\pi_2 = \dots, -3, \mathbf{0}, -2, -1, 1, 2, 3\dots$$

$$\pi_3 = \dots, \mathbf{0}, -3, -2, -1, 1, 2, 3\dots$$

$$\vdots$$

gives us an object in which 0 is inverted with all negative integers.
We might expect this to look like

$$0, \dots, -3, -2, -1, 1, 2, 3, \dots$$

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gives us an object in which 0 is inverted with all negative integers.
We might expect this to look like

$$0, \dots, -3, -2, -1, 1, 2, 3, \dots$$

This isn't a permutation! But it is an ordering of the integers.

Definition

Let $\triangleleft$ be a total order of $\mathbb{Z}$. An **inversion** of $\triangleleft$ (with respect to $<$) is a pair (i, j) such that

$$i < j$$

$$i \triangleright j$$

If we associate to a permutation of $\mathbb{Z}$ the total ordering

$$\dots \triangleleft \pi(0) \triangleleft \pi(1) \triangleleft \pi(2) \triangleleft \dots$$

this coincides with inversions for the permutation.

Weak order for total orders?

Let $\text{Tot}(\mathbb{Z})$ be the set of all total orders on $\mathbb{Z}$.

Definition

The **inversion set** for a total order $\triangleleft$ (with respect to $<$) is

$$\text{Inv}(\triangleleft) = \{(i, j) \mid i < j, i \triangleright j\}.$$

The **weak order** on $\text{Tot}(\mathbb{Z})$ is given by containment of inversion sets.

Theorem

The weak order on $\text{Tot}(\mathbb{Z})$ is a complete lattice.

How do total orders relate to biclosed sets?

Theorem (B–Speyer 22)

The map

$$\begin{aligned}\mathrm{Tot}(\mathbb{Z}) &\rightarrow 2^{\Phi^+} \\ \triangleleft &\mapsto \{\mathbf{e}_i - \mathbf{e}_j \mid (i, j) \in \mathrm{Inv}(\triangleleft)\}\end{aligned}$$

is an order isomorphism from $\mathrm{Tot}(\mathbb{Z})$ to $\mathrm{Bic}(S_\infty)$.

Affine permutations

Definition

The **affine symmetric group** $\tilde{S}_n$ is the group of bijections $\tilde{\pi} : \mathbb{Z} \rightarrow \mathbb{Z}$ satisfying:

(a) $\tilde{\pi}(i + n) = \tilde{\pi}(i) + n$ for all $i \in \mathbb{Z}$, and

(b)
$$\sum_{i=1}^n \tilde{\pi}(i) = \sum_{i=1}^n i.$$

$\tilde{S}_n$ is the Coxeter group of type $\tilde{A}_{n-1}$.

The type $\tilde{A}_{n-1}$ root system

Let $\tilde{V}$ be the vector space spanned by vectors $\tilde{\alpha}_{i,j}$ with $i, j \in \mathbb{Z}$, subject to the relations

$$\tilde{\alpha}_{i,j} + \tilde{\alpha}_{j,k} = \tilde{\alpha}_{i,k}$$

and

$$\tilde{\alpha}_{i,j} = \tilde{\alpha}_{i+n,j+n}$$

for all $i, j, k \in \mathbb{Z}$.

The root system of $\tilde{S}_n$ is

$$\tilde{\Phi} = \{\tilde{\alpha}_{i,j} \mid i \not\equiv j \pmod{n}\}.$$

The biclosed sets for $\tilde{A}_{n-1}$

Given a total order $\triangleleft$ on $\mathbb{Z}$, define

$$I(\triangleleft) = \{\tilde{\alpha}_{i,j} \mid (i,j) \in \text{Inv}(\triangleleft), i \not\equiv j \pmod{n}\}.$$

We say $\triangleleft$ is **translation invariant** if, for all $i, j \in \mathbb{Z}$,

$$i \triangleleft j \quad \text{iff} \quad i + n \triangleleft j + n.$$

Theorem (B–Speyer 2022)

A set $B \subseteq \tilde{\Phi}^+$ is biclosed if and only if there is some translation invariant total order $\triangleleft$ such that $B = I(\triangleleft)$.

This leads to a very explicit parametrization of $\text{Bic}(\tilde{A}_{n-1})$, using **window notation**.

Applications of biclosed sets

Application 1: Coxeter matroid polytopes

Let W be a Coxeter group with root system $\Phi \subseteq V$

Definition

Let $\rho \in V$ be a vector not fixed by reflection over any root. A finite subset $X \subseteq W$ is a **Coxeter matroid** if every edge of the polytope

$$\text{Pol}_X = \text{conv}(\{x\rho \mid x \in X\})$$

is parallel to a root.

Let G_X be the 1-skeleton of Pol_X , with edges directed in the negative root direction.

Conjecture

For any Coxeter matroid X and biclosed set $B \subseteq \Phi^+$, reverse the edges of G_X which are parallel to roots in B . Then the resulting directed graph is acyclic and has a unique source and sink.

Application 2: Bruhat decompositions

Conjecture

For any Coxeter matroid X and biclosed set $B \subseteq \Phi^+$, reverse the edges of G_X which are parallel to roots in B . Then the resulting directed graph is acyclic and has a unique source and sink.

Corollary

For a Kac–Moody group G with Borel subgroup B , and any biclosed set $A \subseteq \Phi^+$, there is a “twisted Bruhat decomposition”

$$G = \bigsqcup_{w \in W} U_w wB$$

where U_w is a product of root subgroups depending on A and w .

Application 2: Bruhat decompositions

Conjecture

For any Kac–Moody algebra $\mathfrak{g}$ with roots Φ , and any biclosed set $A \subseteq \Phi^+$, there is a subalgebra $\mathfrak{a} \subseteq \mathfrak{g}$ such that the real root space $\mathfrak{g}_\alpha \subseteq \mathfrak{a}$ if and only if $\alpha \in A$ or $-\alpha \in \Phi^+ \setminus A$.

This would mean biclosed sets are exactly the positive roots appearing in some generalized Borel subalgebra.

Corollary

The twisted Bruhat decomposition

$$G = \bigsqcup_{w \in W} U_w wB$$

is the decomposition into orbits of a generalized Borel subgroup of G acting on G/B .

Application 3: Cluster algebras

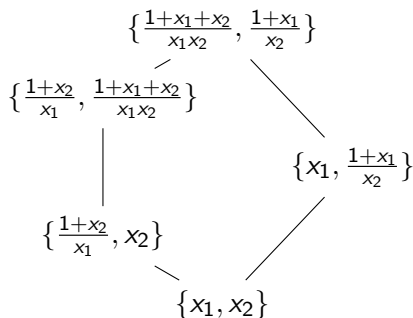
Associated to each (loopless 2-acyclic) quiver Q is a **cluster algebra** A_Q .

$$Q = \bullet \longrightarrow \bullet$$

Each node has a variable x_i attached

Clusters are sets of Laurent polynomials in $x_1, \dots, x_n$ built from $\{x_1, \dots, x_n\}$ using **mutation**

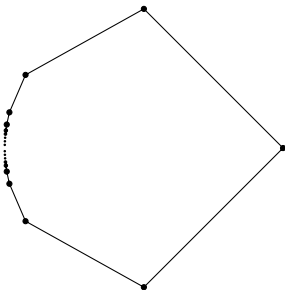
The **exchange graph** has vertices the clusters and edges the mutations



Application 3: Cluster algebras

Now consider the “type $\tilde{A}_1$ ” quiver

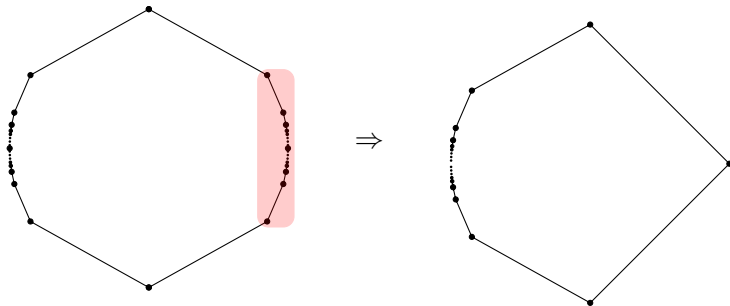
$$Q = \bullet \begin{matrix} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{matrix} \bullet$$



Application 3: Cluster algebras

Now consider the “type $\tilde{A}_1$ ” quiver

$$Q = \bullet \rightrightarrows \bullet$$



Application 3: Cluster algebras

Theorem (B–Defant 2024, B 2025)

If Q is any orientation of an affine Dynkin diagram with affine Coxeter group W , then there is a quotient of the extended weak order of W which realizes the ordered exchange graph of A_Q .

These are called **affine Cambrian quotients**

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Conjecture

If Q is any acyclic quiver, and one interprets Q as a Coxeter–Dynkin diagram with Weyl group W , then there is a quotient of the extended weak order of W which realizes the ordered exchange graph of A_Q .

Application 4: Torsion classes

A **torsion class** for an algebra Π is a collection of finite-dimensional Π -modules which is closed under quotients and extensions.
Torsion classes form a lattice under containment.

Theorem (B 2025)

If W is a (simply-laced) affine Coxeter group, then the extended weak order of W is a quotient of the lattice of torsion classes of a **preprojective algebra**.

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Conjecture

If W is any (crystallographic) Coxeter group, then the extended weak order of W is a quotient of the lattice of torsion classes of a preprojective algebra Π_W .

Thank you!

Affine permutations

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Condition (b) lets us recover $\tilde{\pi}$ from one-line notation: e.g. the identity is

$$\dots, -3, -2, -1, 0, 1, 2, 3, 4, 5, \dots$$

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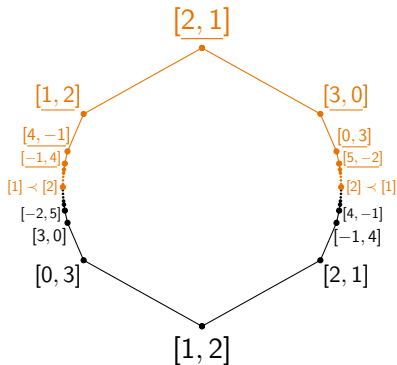
$$\dots, -3, -2, -1, 0, 1, 2, 3, 4, 5, \dots$$

We can also encode an affine permutation via **window notation**:
e.g. in $\tilde{S}_3$

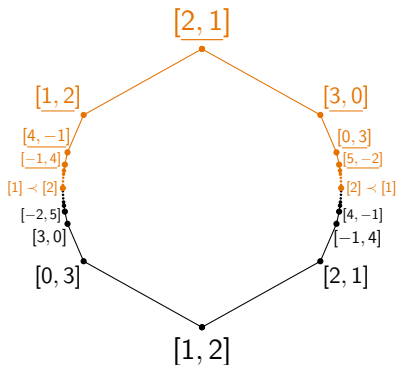
$$[0, 1, 2] = [1, 2, 3] = \text{id},$$

$$[1, 0, 2] = \dots, -2, -3, -1, 1, 0, 2, 4, 3, 5, \dots$$

Extended weak order



Extended weak order



The black vertices are labeled with elements of $\widetilde{S}_n$.
What are the other labels?

Definition (B.–Speyer 2022)

A **translation invariant total order (TITO)** is a total order ($\prec$) on $\mathbb{Z}$ satisfying the following property:

- For all $i, j \in \mathbb{Z}$, we have $i \prec j$ if and only if $i + n \prec j + n$.

A TITO is **real** if it further satisfies:

- For all $i \in \mathbb{Z}$, if $i + n \prec i$ then there exists a k with $i + n \prec k \prec i$.

TITOs can be encoded with window notation: e.g. in $\tilde{\mathcal{B}}_3$ the notation $[1] \prec [3, 2]$ encodes

$$\dots \prec -2 \prec 1 \prec 4 \prec 7 \prec \dots \prec \dots \prec 0 \prec -1 \prec 3 \prec 2 \prec 6 \prec 5 \prec \dots$$

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We write $[1] \prec [3, 2]$ to instead encode

$$\dots \prec -2 \prec 1 \prec 4 \prec 7 \prec \dots \prec \dots \prec 6 \prec 5 \prec 3 \prec 2 \prec 0 \prec -1 \prec \dots$$

Theorem (B.–Speyer 2022)

The map $\triangleleft \mapsto I(\triangleleft)$ is a bijection from real TITOs to $\text{Bic}(\tilde{A}_{n-1})$.

An example

Consider the sequence in $\tilde{S}_2$:

$$\begin{array}{ccccccccccccccccccc} \dots & \prec & -5 & \prec & -4 & \prec & -3 & \prec & -2 & \prec & -1 & \prec & 0 & \prec & 1 & \prec & 2 & \prec & 3 & \prec & 4 & \prec & 5 & \prec & \dots \\ \dots & \prec & -5 & \prec & -2 & \prec & -3 & \prec & 0 & \prec & -1 & \prec & 2 & \prec & 1 & \prec & 4 & \prec & 3 & \prec & 6 & \prec & 5 & \prec & \dots \\ \dots & \prec & -5 & \prec & 0 & \prec & -3 & \prec & 2 & \prec & -1 & \prec & 4 & \prec & 1 & \prec & 6 & \prec & 3 & \prec & 8 & \prec & 5 & \prec & \dots \\ & \vdots \end{array}$$

The join in extended weak order is

$$\dots \prec -4 \prec -2 \prec 0 \prec 2 \prec 4 \prec \dots \prec -3 \prec -1 \prec 1 \prec 3 \prec 5 \prec \dots$$

An example

